

Positive Square Energy for Two Pentagons Joined by an Arbitrary Path

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Abstract

We prove a uniform positive-square-energy bound for every connected cactus whose only cyclic blocks are two vertex-disjoint pentagons. The pentagons may be joined by a path of any positive length, and arbitrary trees may be attached anywhere. If the graph has order n , then

$$s^+(G) > n + 5 - 2\sqrt{5} > n.$$

The proof does not compare the graph with its bare bicyclic core: attached trees can increase that core's pointwise Coulson phase. Instead, the joining path is factored exactly, and the phase of the joined graph is compared with the sum of the phases of the two actual unicyclic lobes.

1 Statement and notation

All graphs are finite, simple, and undirected. If the adjacency eigenvalues of a graph H are $\lambda_1, \dots, \lambda_h$, put

$$s^+(H) = \sum_{\lambda_j > 0} \lambda_j^2, \quad s^-(H) = \sum_{\lambda_j < 0} \lambda_j^2, \quad D(H) = s^+(H) - s^-(H).$$

Theorem 1.1. *Let G be a connected cactus whose only cyclic blocks are two copies of C_5 . Suppose that the two cycles are vertex-disjoint and that the unique path joining them has positive length. Arbitrary trees may be attached at any vertices of the cycles or the joining path. If $n = |V(G)|$, then*

$$s^+(G) > n + 5 - 2\sqrt{5} > n. \tag{1}$$

Thus the theorem includes a single bridge between the cycles and every longer joining path. It does not include the bouquet in which the two pentagons share a vertex; that graph has no positive-length path with distinct cycle endpoints and requires a separate theorem.

2 Normalized characteristic polynomials

For a graph F , define its signless matching partition by

$$Z_F(t) = \sum_M t^{|V(F)| - 2|M|}, \tag{2}$$

where the sum is over all matchings. More generally, for positive vertex activities $a = (a_v)$, write

$$Z_F(a) = \sum_M \prod_{v \text{ unmatched by } M} a_v. \tag{3}$$

All edge activities are one. Every such partition is positive.

For a graph H of order h , let

$$\Psi_H(t) = i^{-h} \det(itI - A(H)) = \prod_{j=1}^h (t + i\lambda_j), \quad t > 0. \quad (4)$$

Grouping Sachs' expansion by its collection $\mathcal{C}$ of pairwise vertex-disjoint cycle components gives

$$\Psi_H(t) = \sum_{\mathcal{C}} \prod_{C \in \mathcal{C}} (-2i^{-|C|}) Z_{H-V(C)}(t). \quad (5)$$

Indeed, after the cycles are fixed, the remaining elementary components are a matching; substitution of it changes its alternating matching polynomial into (2). In particular, a pentagon has multiplier

$$-2i^{-5} = 2i. \quad (6)$$

We use the continuous argument

$$\Theta_H(t) = \sum_{j=1}^h \arctan \frac{\lambda_j}{t}, \quad -\frac{\pi}{2} < \arctan x < \frac{\pi}{2}. \quad (7)$$

It is the continuous argument of (4) that tends to zero as $t \rightarrow \infty$.

Lemma 2.1 (Signed Coulson identity). *For every graph H ,*

$$D(H) = -\frac{4}{\pi} \int_0^\infty t \Theta_H(t) dt. \quad (8)$$

The integral is absolutely convergent.

Proof. For every real λ ,

$$\lambda|\lambda| = \frac{2}{\pi} \int_0^\infty \frac{\lambda^3}{\lambda^2 + t^2} dt.$$

Also

$$\Theta'_H(t) = -\sum_j \frac{\lambda_j}{\lambda_j^2 + t^2}, \quad \sum_j \frac{\lambda_j^3}{\lambda_j^2 + t^2} = -t^2 \sum_j \frac{\lambda_j}{\lambda_j^2 + t^2},$$

where $\sum_j \lambda_j = \text{tr } A(H) = 0$ was used in the second identity. Summing the scalar formula and integrating by parts therefore gives (8). At zero, $t^2 \Theta_H(t) \rightarrow 0$. At infinity, $\sum_j \lambda_j = 0$ and the arctangent expansion give $\Theta_H(t) = O(t^{-3})$. These estimates also prove absolute convergence. $\square$

3 Tree splitting and the two actual lobes

Let the joining path have endpoints u_1 and u_2 on the two pentagons and let it have $m \geq 0$ internal vertices. Thus $m = 0$ is the bridge case. Put into X_i the i th pentagon together with every tree branch attached at its cycle vertices, and root X_i at u_i . The graphs X_1, X_2 are the *actual* pentagonal unicyclic lobes; no cycle attachment is discarded. Only tree branches attached to internal vertices of the joining path remain to be eliminated.

We record the elementary matching belief-propagation split. Orient a rooted tree branch toward the path. If $x \rightarrow p$ is an oriented edge and $T_{x \rightarrow p}$ is the component below x , including x , set

$$q_{x \rightarrow p}(t) = \frac{Z_{T_{x \rightarrow p}}(t)}{Z_{T_{x \rightarrow p} - x}(t)}. \quad (9)$$

Splitting a matching according to the status of x gives

$$q_{x \rightarrow p}(t) = t + \sum_{y \text{ child of } x} \frac{1}{q_{y \rightarrow x}(t)} \geq t > 0. \quad (10)$$

After all path-internal branches are split off, the j th internal path vertex has a positive effective activity

$$a_j(t) = t + \sum_{x \text{ attached at } j} \frac{1}{q_{x \rightarrow j}(t)} > 0, \quad (11)$$

and all terms have a common positive real prefactor

$$B_0(t) = \prod_{x \rightarrow j \text{ at an internal path vertex}} Z_{T_{x \rightarrow j}}(t) > 0. \quad (12)$$

To verify the common factor even when a vertex is removed by a matching edge, note that extracting Z_T gives the edge into a branch the weight $Z_{T-x}/Z_T = 1/q_{x \rightarrow j}$. If the path vertex is not retained, the detached branch contributes its full factor Z_T . Thus no hidden quotient remains.

Define

$$z_i = \Psi_{X_i}(t), \quad w_i = \Psi_{X_i - u_i}(t), \quad r_i = \frac{w_i}{z_i}. \quad (13)$$

The graph $X_i - u_i$ is a forest, so $w_i = Z_{X_i - u_i}(t) > 0$. Formula (5) applied to the unique pentagon of X_i gives

$$z_i = Z_{X_i}(t) + 2iZ_{X_i - V(C_5)}(t). \quad (14)$$

Both displayed matching partitions are positive. Hence z_i is in the open first quadrant. Writing

$$r_i = x_i - iy_i \quad (15)$$

therefore gives $x_i > 0$ and $y_i > 0$: each r_i is in the open fourth quadrant. Put $\theta_i = \text{Arg } z_i \in (0, \pi/2)$.

4 The exact connector factor

For a list of variables define the continuant by

$$K() = 1, \quad K(b_1) = b_1, \quad K(b_1, \dots, b_k) = b_k K(b_1, \dots, b_{k-1}) + K(b_1, \dots, b_{k-2}). \quad (16)$$

Proposition 4.1 (Connector factor). *With the preceding notation,*

$$\Psi_G(t) = B_0(t) z_1 z_2 F(t), \quad (17)$$

where

$$F = \begin{cases} 1 + r_1 r_2, & m = 0, \\ a_1 + r_1 + r_2, & m = 1, \\ K(a_1 + r_1, a_2, \dots, a_{m-1}, a_m + r_2), & m \geq 2. \end{cases} \quad (18)$$

Moreover, $\text{Im } F < 0$.

Proof. First ignore the already extracted factor B_0 . In each grouped Sachs term, split matchings along the joining path. Summing over the two possible Sachs states inside a lobe gives z_i when its boundary edge is unused; using that edge deletes the root and gives w_i . Thus $w_i/z_i = r_i$ records use of the boundary edge at the i th lobe. For $m = 0$, the bridge is either unused or used, giving $1 + r_1 r_2$. For $m = 1$, the internal vertex is unmatched, or is matched to exactly one boundary, giving $a_1 + r_1 + r_2$. For $m \geq 2$, the usual last-edge split for path matchings is exactly recurrence (16), with the two boundary contributions added to the first and last activities. This proves (17)–(18).

For the sign of the imaginary part, expand

$$F = A + Br_1 + Cr_2 + Dr_1 r_2. \quad (19)$$

Here B is the coefficient of r_1 and C is the coefficient of r_2 . For $m = 0$ one has $(A, B, C, D) = (1, 0, 0, 1)$, while for $m = 1$ one has $(A, B, C, D) = (a_1, 1, 1, 0)$. If $m \geq 2$, the matching expansion of the continuant shows that all four coefficients are positive: A, B, C, D are the positive partition sums obtained by prescribing neither, only the left, only the right, or both boundary edges, respectively. These four classes are nonempty for every $m \geq 2$.

Using (15),

$$\text{Im } F = -By_1 - Cy_2 - D(x_1 y_2 + x_2 y_1) < 0. \quad (20)$$

For $m = 0$ the last term is strictly negative; for $m = 1$ the middle two terms are strictly negative; and for $m \geq 2$ all three displayed contributions are negative. This proves the assertion in every case. $\square$

The identities in Proposition 4.1 are independently corroborated by the exact symbolic and random-cactus audit script

`positive-square-energy/experiments/connector_factor_audit.py`.

The script is not needed for the proof.

5 Strict phase comparison

Proposition 5.1. *For every $t > 0$,*

$$0 < \Theta_G(t) < \theta_1(t) + \theta_2(t). \quad (21)$$

Consequently,

$$D(G) > D(X_1) + D(X_2). \quad (22)$$

Proof. Because the only cycles of G are its two pentagons, the imaginary part of the grouped Sachs expansion (5) is exactly

$$\text{Im } \Psi_G(t) = 2Z_{G-V(C_5^{(1)})}(t) + 2Z_{G-V(C_5^{(2)})}(t) > 0. \quad (23)$$

The empty-cycle and double-cycle terms are real. Thus $\Psi_G(t)$ lies in the open upper half-plane, and its continuous argument satisfies $0 < \Theta_G(t) < \pi$.

By Proposition 4.1, the principal argument $\varphi = \text{Arg } F$ belongs to $(-\pi, 0)$. Since $\theta_1 + \theta_2 \in (0, \pi)$, the number $\theta_1 + \theta_2 + \varphi$ lies in $(-\pi, \pi)$ and is congruent modulo 2π to Θ_G . It cannot be nonpositive: adding 2π would then put the corresponding argument in $(\pi, 2\pi)$, contradicting $0 < \Theta_G < \pi$. Therefore

$$\Theta_G = \theta_1 + \theta_2 + \text{Arg } F, \quad (24)$$

and (21) follows strictly from $\text{Arg } F < 0$.

Apply the Coulson identity to G, X_1, X_2 . All three phase integrals converge, and the integrand $t(\theta_1 + \theta_2 - \Theta_G)$ is continuous and strictly positive on $(0, \infty)$. Hence

$$D(G) - D(X_1) - D(X_2) = \frac{4}{\pi} \int_0^\infty t(\theta_1 + \theta_2 - \Theta_G) dt > 0,$$

which is (22). $\square$

This is deliberately not a comparison with the phase of the bare two-cycle core. Such a comparison is false in general: attachments may increase the bare-core phase pointwise. The valid comparison is with the sum of the phases of the actual lobes X_1 and X_2 , after which the connector supplies the strictly negative phase $\text{Arg } F$.

6 The pentagonal lobe bound

Lemma 6.1. *If X is a unicyclic graph whose unique cycle is C_5 , then*

$$D(X) \geq D(C_5) = 4 - 2\sqrt{5}. \quad (25)$$

Proof. Apply the same branch split (9)–(10) to all trees attached to the five cycle vertices. It gives a common positive factor and effective cycle activities $b_v = t + y_v \geq t$. Formula (5) becomes

$$\Psi_X(t) = K_0(t)(Z_{C_5}(b) + 2i), \quad K_0(t) > 0. \quad (26)$$

The weighted matching partition has nonnegative coefficients in all vertex activities, so

$$Z_{C_5}(b) \geq Z_{C_5}(t). \quad (27)$$

Both quantities in (26) lie in the first quadrant. Consequently

$$0 < \Theta_X(t) = \arctan \frac{2}{Z_{C_5}(b)} \leq \arctan \frac{2}{Z_{C_5}(t)} = \Theta_{C_5}(t).$$

The Coulson identity yields $D(X) \geq D(C_5)$.

For completeness, the eigenvalues of C_5 are $2\cos(2\pi j/5)$ for $0 \leq j < 5$. Its positive eigenvalues are 2 and $(\sqrt{5} - 1)/2$ with multiplicity two; hence

$$s^+(C_5) = 4 + 2 \left(\frac{\sqrt{5} - 1}{2} \right)^2 = 7 - \sqrt{5}.$$

Since C_5 has five edges, $s^+(C_5) + s^-(C_5) = \text{tr } A(C_5)^2 = 10$, and therefore $D(C_5) = 2s^+(C_5) - 10 = 4 - 2\sqrt{5}$. $\square$

Proof of Theorem 1.1. Proposition 5.1 and Lemma 6.1 give

$$D(G) > 2(4 - 2\sqrt{5}) = 8 - 4\sqrt{5}. \quad (28)$$

The connected graph G has cycle rank two, so if $e = |E(G)|$, then $e = n + 1$. Therefore

$$s^+(G) + s^-(G) = \text{tr } A(G)^2 = 2e = 2n + 2.$$

Averaging this identity with (28) proves

$$s^+(G) = n + 1 + \frac{D(G)}{2} > n + 1 + 4 - 2\sqrt{5} = n + 5 - 2\sqrt{5}.$$

Finally, $5 - 2\sqrt{5} > 0$ because $25 > 20$, completing the proof. $\square$